

Md Omar Faruqe RIYAD
Software Engineer

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 EDUCATION

2018–2023 **B.Sc. in Computer Science and Engineering**
Shahjalal University of Science and Technology (SUST)
CGPA : **3.53/4.00** (Last two years : **3.83**)

★ STANDARDIZED TEST SCORES

GRE : 317/340 (Quantitative : 160, Verbal : 157, Analytical Writing : 3)
TOEFL : 104/120 (Reading : 29, Listening : 27, Speaking : 23, Writing : 25)

 RESEARCH EXPERIENCE

Shared Tasks	
2025	Code Generation in Bangla – BLP Workshop, IJCNLP-AAACL 2025 - Investigated low-resource language code generation by fine-tuning open-source models (Llama 3, TigerLLM, Qwen 2.5 Coder, Qwen 3) using parameter-efficient LoRA adapters. - Enhanced model reasoning via Chain-of-Thought (CoT) augmented fine-tuning and a self-refinement loop that iteratively critiques and corrects generated code based on execution feedback. - Conducted rigorous error analysis to identify and mitigate failure modes like instruction misinterpretation and overfitting, improving generalization and achieving 85% accuracy.
2024	BRAINTEASER : A Novel Task Defying Common Sense - SemEval 2024 Designed data augmentation pipelines to improve the robustness of the model for common sense reasoning tasks and conducted a comparative analysis on advanced language models.
2023	Violence Inciting Text Detection (VITD) – BLP Workshop at EMNLP 2023 - Applied semi-supervised self-training techniques to address the small size of the data set and class imbalance, utilizing external data sets to improve model robustness. - Enhanced dataset diversity and quality through back-translation, improving semantic variety, and correcting linguistic inconsistencies. - Implemented an ensemble approach with bagging and majority voting to reduce prediction variance.
Undergraduate Thesis	
2022	Improving Answer Space Diversity in Visual Question Answering (VQA) through Automatic Question and Answer Generation - Conducted a comparative study of various VQA methods, analyzing their limitations - Addressed Answer Space Diversity limitation in VQA v2 dataset by augmenting training data with automatically generated, contextually relevant question-answer pairs.

 PUBLICATIONS

Deja Vu at SemEval 2024 Task 9 : A Comparative Study of Advanced Language Models for Commonsense Reasoning
Accepted in *The SemEval 2024, NAACL, Mexico*

Team_Syrax at BLP-2023 Task 1 : Data Augmentation and Ensemble Based Approach for Violence Inciting Text Detection in Bangla
Accepted and Presented at *The BLP 2023, EMNLP, Singapore*

AdversaryAI at BLP-2025 Task 2 : A Think, Refine, and Generate (TriGen) System with LoRA and Self-Refinement for Code Generation
Under Review at *The BLP 2025, IJCNLP-AAACL*

 SKILLS

Languages	Java, C/C++, Python, JavaScript, SQL, Shell
Frameworks & Libraries	JEE (JSF, CDI, EJB, JPA, JMS with OpenMQ), GlassFish, PrimeFaces, Flask, FastApi, React
Databases & Data Storage	PostgreSQL, MySQL, Neo4j (Graph), MinIO (S3-compatible)
DevOps & Observability	Docker, Git, GitHub Actions CI/CD, Ansible, Prometheus, Grafana, Loki
AI & ML	PyTorch, LangChain, RAG, LLM, Vector Database (ChromaDB), OpenAI & Ollama APIs
Security	Jakarta Security, JWT, Foundational Cryptography
Tools	Flyway, JasperReports, $\LaTeX$

JOB EXPERIENCE

Present
Oct 2025

AI Engineer, ALTRI.AI, US (part-time)

- > ML and LLM-based hybrid scoring pipeline with explainable predictions.
- > Feature extraction based on Image and Text analysis.

Python RAG LLM FastApi React PostgreSQL

Present
April 2023

Software Engineer, DYNAMIC SOLUTION INNOVATORS, Bangladesh

- > Engineered security critical enterprise applications for government infrastructures, translating complex offline bureaucratic processes into streamlined digital workflows.
- > Served as a key member of the internal cybersecurity team, conducting preliminary security testing.

Key Projects :

E-Appeal (NBR • USAID)

- > Pioneered the company's first large-scale JSF application, establishing engineering guidelines.
- > Integrated Message Queue for **asynchronous, decoupled event handling**.

Project Scratch

Internal reusable Core Services Platform to accelerate development.

- > Engineered centralized **observability stack** with Prometheus, Loki, and Grafana.
- > **Dockerized the complete application stack**, including GlassFish, PostgreSQL, MinIO, and RabbitMQ.
- > Developed a reusable Lazy Data Model component and **contributed a related feature enhancement to the official PrimeFaces open-source library**. (PR #12865)

SHMS (Smart Hotel Management System)

Nationwide licensing and registration system for hotels and restaurants

- > Contributed to the design and implementation of a **state-driven approval workflow**, reusable across multiple application types, minimizing code duplication
- > Utilized **AOP** to cleanly separate cross-cutting concerns like notifications from core business logic.

BCIC-ERP

End-to-end Integrated ERP system for process automation across 12 independent modules

- > **Assumed full ownership** of the Procurement module, leading the process from business analysis (As-Is/To-Be modeling) and database schema design to hands-on development and successful client delivery.

Java JEE JSF Primefaces CDI EJB JPA PostgreSQL JMS Glassfish Prometheus Grafana Loki

PROJECTS AND OPEN SOURCE CONTRIBUTION

KNOWLEDGERELAY : AI-BASED KNOWLEDGE TRANSFER AGENT

2025

 github.com/riyadomf/knowledgeRelay

Built a RAG system to preserve project-specific knowledge base and assist new team members via contextual Q&A.

- > Engineered a multi-stage knowledge ingestion pipeline that automatically generates question-answer pairs from documents to fine-tune the retrieval process.
- > Integrated ChromaDB with vector embeddings and metadata-based filtering for source grounded efficient retrieval.
- > Implemented a contextualize_qa_chain to rephrase user queries using chat history for more accurate retrieval.

Python FastAPI LangChain ChromaDB OpenAI Ollama React

PRIMEFACES : JPALAZYDATAMODEL FEATURE ENHANCEMENT

2024

 github.com/primefaces/primefaces/pull/12865

Contributed to the PrimeFaces open-source project by adding the ability to inject custom FilterMeta into the JPALazyData-Model component. This enhancement allows developers to perform advanced, customizable filtering within DataTables.

Java JSF Primefaces JPA

SHOPAHOLIC

2022

 github.com/Nowshadjunaed/Shopaholic

Built a full-stack e-commerce platform connecting customers, banks, and suppliers.

JavaScript Node.js Express.js MongoDB React Redux

BREVITY

2021

 github.com/riyadomf/Brevity

Developed a responsive blog platform designed to connect beginners with experts in various fields. The platform enables experienced individuals to share knowledge, resources, and help others with learning difficulties.

Python Flask MySQL JavaScript HTML CSS

TEACHING EXPERIENCE

Present
2024

Course Instructor, **DYNAMIC LEARNING**,

- > Authored and instructed a course - Enterprise Web Development with JEE, JSF, and PrimeFaces. 
- > Demonstrated core architectural patterns including component-driven design, CDI/EJB integration, backing bean scopes, and JSF lifecycle and state management.

PROBLEM SOLVING

Solved over 500+ problems with C/C++ in different online judges

- > Vjudge:  [riyad2017331042](#) | LeetCode:  [riyad42](#) | LightOJ:  [riyadomf](#)
- > StopStalk Profile:  [riyadomf](#)

CERTIFICATIONS & ACHIEVEMENTS

- > 4th Place out of 32 teams – Code Generation in Bangla, A shared task at the BLP Workshop, IJCNLP-AAACL 2025
- > 6th Place out of 30 teams – DSI AI Agent Hackathon, 2025
- > ISC2 Certified in Cybersecurity (CC)
- > Top 20 (Rank 19 → 12 post-eval) – Violence Inciting Text Detection (VITD), BLP 2023

EXTRA-CURRICULARS

- > Volunteer, IEEE International Conference on Bangla Speech and Language Processing (ICBSLP), 2019.
- > Ranked 11th in BUET CSE Fest 2022 AI Contest and 19th in Battle of Brains by Green University
- > Runners-Up, Intra-Department Football Tournament, 2022.